A PAIR IS HENSELIAN EXACTLY WHEN THE PAIR MODULO AN IDEAL AND THE QUOTIENT PAIR ARE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The henselian property is equivalent to bijectivity on idempotents for every integral algebra. Lemma \ref{lemma-characterize-henselian-pair}, cited at proof lines 2777 and 2784 and 2792-2795. Is the criterion applied to the two nested pairs and to the chain of quotient maps.
- The henselian property is preserved by integral ring maps. Lemma \ref{lemma-integral-over-henselian-pair}, cited at proof lines 2785-2786. Gives the henselian property of the quotient pair from the larger pair.

Source reference. Original lemma 0DYD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17463–17463. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 09XI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12531–12531. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 09XK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12533–12533. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $I \subset J \subset A$ be ideals of a ring A. The following are equivalent*

(1) *(A, I) and $(A/I, J/I)$ are henselian pairs, and*

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(2) (A, J) *is an henselian pair.*

Proof. Assume (1). Let B be an integral A-algebra. Consider the ring maps

$$B \to B/IB \to B/JB$$

By Lemma 09XI we find that both arrows induce bijections on idempotents. Hence so does the composition. Whence (A, J) is a henselian pair by Lemma 09XI.

Conversely, assume (2) holds. Then $(A/I, J/I)$ is a henselian pair by Lemma 09XK. Let B be an integral A-algebra. Consider the ring maps

$$B \to B/IB \to B/JB$$

By Lemma 09XI we find that the composition and the second arrow induce bijections on idempotents. Hence so does the first arrow. It follows that (A, I) is a henselian pair (by the lemma again). □